$j = \infty$ (computational)
q_0
q_1
$j = 0$
q_0
H
q_1
H
$j = 1$
q_0
H
q_1
H
S
$j = 2$
q_0
H
S
q_1
H
$j = 3$
q_0
H
S
q_1
H
S